14.1 link margin required for L-band communication: 20~25dB in vehicles, 30~40dB in buildings. with 16 dB margin, the Iridium system can't reliably work in these environments, as this margin isn't enough to offset the heavy fading from obstacles.

14.2 ① main beam position.

求 $F_\theta(\theta,\phi)_{max}$, $F_\theta(\theta,\phi) = A\sin^2\theta\cos\phi$

when $\theta=\frac{\pi}{2}$, ~~cos~~ $\phi=0$, $F_\theta(\theta,\phi)_{max}=A$.

则 main beam position is at $\theta=\frac{\pi}{2}$, $\phi=0$.

② 3dB Beamwidths in the Principal planes.

在E-plane中, $\theta=\frac{\pi}{2}$, $F_\theta(\theta,\phi)=F_\theta(\frac{\pi}{2},\phi)=A\cos\phi$

此时 $F_\theta(\theta,\phi)_{max}=A$.

当 $F_\theta(\theta,\phi)=\frac{1}{\sqrt{2}}A$ 时, $\phi=\pm\frac{\pi}{4}$, 则 $\Delta\phi=\frac{\pi}{2}$

在H-plane中, $\phi=0$, $F_\theta(\theta,\phi)=A\sin^2\theta$.

此时 $F_\theta(\theta,\phi)_{max}=A$.

当 $F_\theta(\theta,\phi)=\frac{1}{\sqrt{2}}A$ 时, $\sin\theta=2^{-\frac{1}{4}}$

$\theta=\arcsin\theta=0.99$ radians

则 $\Delta\theta=\pi-0.99\times2=1.14$ radians

③ Directivity (in dB)

$$D=\frac{F_{\theta max}^2}{F_{\theta average}^2}, \quad F_{\theta max}^2=A^2$$

$$F_\theta^2 average=\frac{1}{4\pi}\int_0^{2\pi}\int_0^{\pi}A^2\sin^4\theta\cos^2\phi\sin\theta\,d\theta\,d\phi$$

$$=\frac{1}{4\pi}\int_0^{2\pi}\cos^2\phi\,d\phi\cdot\int_0^{\pi}\sin^5\theta\cdot d\theta\cdot A^2$$

$$=\frac{4}{15}A^2$$

则 $D=\frac{A^2}{\frac{4}{15}A^2}=\frac{15}{4}$

in dB: $D=10\lg\frac{15}{4}=5.7dB$.

④ 函数中仅含cosφ, 则天线为线极化(垂直极化).

linearly polarized (vertical)

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14.3 $F_\theta(\theta,\phi) = \begin{cases} A\sin\theta, & 0 \le \theta \le \frac{\pi}{2} \\ 0, & \text{其他} \end{cases}$.

$$D = \frac{4\pi F_{max}^2}{\int_\theta^\pi \int_\phi^{2\pi} F_\theta^2(\theta,\phi)\sin\theta\, d\phi\, d\theta} = \frac{4\pi}{\frac{4}{3}\pi} = 3$$

in dB: $D = 10\lg D = 4.8\,dB$

14.4 $d = 18$ inches $= 18 \times 0.0254 = 0.4572$ m

$f = 12.4$ GHz, 则 $\lambda = \frac{c}{f} = 0.0242$ m.

$A = \pi R^2 = \pi \cdot \left(\frac{d}{2}\right)^2$

$$D = \frac{4\pi A}{\lambda^2} \cdot \eta_{rad} = \frac{4\pi \times \pi \left(\frac{0.4572}{2}\right)^2}{0.0242} \times 65\% = 2287$$

in dB, $D_{dB} = 10\lg D = 33.6\,dB$.

14.5 (a). Apertnre Efficiency.

$D = \frac{G}{\eta_{rad}} = 10^{3.9}/0.9 = 8826$, $d = 12$ inches $\times 0.0254$ m/inch $= 0.3048$ m.

$D = \frac{4\pi A}{\lambda^2} \cdot \eta_{rad}$, 则 $\eta_{rad} = \frac{\lambda^2 D}{4\pi A} = \left(\frac{c}{\lambda f d}\right)^2 \cdot D = 60\%$

(b) $D = \frac{32400}{\theta_1 \theta_2} = \frac{32400}{\theta_3^2}$

则 $\theta_3 = \sqrt{\frac{32400}{8826}} = 1.9°$

14.6 $T_A = \eta_{rad} T_b + (1-\eta_{rad}) T_p$

则 $\eta_{rad} = \frac{T_A - T_p}{T_b - T_p} = \frac{105-290}{5-290} = 65\%$

14.7. $T_A = \eta_{rad} T_b + (1-\eta_{rad}) T_p$, $T_L = (L-1) T_p$. ~~$T_s = \frac{T}{L}$~~

$$T_s = \frac{1}{L}(T_A + T_L) = \frac{1}{L}[\eta_{rad} T_b + (1-\eta_{rad}) T_b) + (L-1) T_p)$$

$$= \frac{1}{L}[\eta_{rad} T_b + (1-\eta_{rad}) T_p] + \frac{L-1}{L} \cdot T_p \quad (14.20)$$

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14.8 $T_R = (F-1)T_0 = (1.29-1)\times 290 = 84K$

$\eta_{rad} = \frac{1}{L} = \frac{1}{1.78} = 0.56$

$T = \eta_{rad} T_b + (1-\eta_{rad}) T_p + T_R = 0.56\times 50 + (1-0.56)\times 290 + 84 = 240K$

$\frac{G}{T} = 10\log_{10}\left(\frac{2240}{240}\right) = 9.7 dB/K$

14.9 $G = \frac{S\cdot 4\pi r^2}{P_t} = \frac{7.5\times 10^{-3} W/m^2 \times 4\pi \times (300m)^2}{85W} = 99.79$

dB: $G = 10\lg 99.79 = 20 dB$

14.10 ① Radio link: $P_r = P_t \cdot \frac{G_t G_r \lambda^2}{(4\pi R)^2}$

$\lambda = \frac{c}{f} = \frac{3\times 10^8 m/s}{28\times 10^9 Hz} = 0.0107m$

$P_t = \frac{1\times (4\pi\times 5000)^2}{316\times 316\times (0.0107)^2} = 3.45\times 10^8 W$

总衰减: $10\log_{10}\left(\frac{P_r}{P_t}\right) = 10\lg\left(\frac{1}{3.45\times 10^8}\right) = -85.4 dB$

② Wired link: $\alpha = 0.05\times 0.115 = 0.0057$

电缆衰减: $10\log e^{-2\alpha R} = 10\lg(e^{-2\times 0.0057\times 5000}) = -250 dB$

总: $-250 + 30\times 4 = -130 dB$

$-85.4 dB > -130 dB$.

则 radio link require the smallest transmit power.

14.11 $T_{sys} = T_A + T_R = T_A + (F_R - 1)T_0$, $\lambda = \frac{c}{f} = 0.317m$

$= 450 + (6.31-1)\times 290 = 1989K$.

$N_0 = kT_{sys}B = 1.38\times 10^{-38}\times 1989\times 200\times 10^3 = 5.5\times 10^{-15} W$.

dBm: $N_0 = 10\lg(5.5\times 10^{-15}\times 10^3) = -112.6 dBm$

$S_0 (dBm) = SNR + N_0 + LM = 10 - 112.6 + 30 = -72.6 dBm$.

$S_0 = 10^{-72.6/10}\times 10^{-3} = 5.5\times 10^{-11} W$

$R = \sqrt{\frac{P_t G_t G_r \lambda^2}{(4\pi)^2 S_0}} = \sqrt{\frac{20\times 1\times 1\times (0.317)^2}{(4\pi)^2\times 5.5\times 10^{-11}}} = 15.2km$

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$T_0=290K$.

14.12. $S_i=1\times10^{-16}W$, $G_A=5dB$, $L=25dB$, $G=10dB$, $\frac{C}{N}=32dB$, $T_A=300K$.

$$C=\frac{S_i G_A G}{L}$$

由 $N=kT_e\frac{G}{L}$ 可得 $T_e=\frac{CL}{k\cdot\frac{C}{N}\cdot G}=\frac{S_i\cdot G_A}{k\cdot\frac{C}{N}}$

$$F=1+\frac{T_e}{T_0}-\frac{T_A}{T_0}-\frac{L-1}{G}$$

$$=1+\frac{1\times10^{-16}\times3.16}{1.38\times10^{-23}\times290\times1580}-\frac{300}{290}-\frac{316.2-1}{10}$$

$$=18.4$$

dB: $F=10\lg(18.4)=12.6\,dB$

14.13. ① SNR=0dB，则 $N_0=kT_BB$.

由 $S_0=\frac{P_tG^2\lambda^2}{(4\pi R)^2}$ 得 $R=\sqrt{\frac{P_tG^2\lambda^2}{(4\pi)^2kT_BB}}=\sqrt{\frac{1000\times2.5^2\times4.48^2}{(4\pi)^2\times1.38\times10^{-23}\times4\times4\times10^6}}$

$=1.9\times10^9m$

② SNR=30dB，则 $S_0=10^3N_0$

$$R=\sqrt{\frac{P_tG^2\lambda^2}{(4\pi)^2\times1000N_0}}=6\times10^7m.$$

∴金星与地球距离为 4.2×10^8m.

∴信号无法到达最远行星.

14.14 $L_0=20\lg\left(\frac{4\pi R}{\lambda}\right)=20\lg\left(\frac{4\pi\times1.6\times10^{11}}{0.1307}\right)=263.7dB$

$P_r(dBW)=P_t(dBW)+G_t+G_r-L_0$.

则 $P_r=12.25+27.6-263.7+61.3=-162.6\,dBW$

则 $P_r=10^{-162.6/10}=5.56\times10^{-17}W$

$N_0=kT_{sys}$

$=1.38\times10^{-23}\times13.5=1.86\times10^{-22}W/Hz$

$$R_b=\frac{P_r}{E_b/N_0\times N_0}=\frac{5.56\times10^{-17}}{1.38\times1.863\times10^{-22}}=216\times10^5bps$$

$=216kbps$

故最大数据速率约为 216 kbps.

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14.15 $S=\frac{P_tG_t}{4\pi R_t^2}$, $S_r=\frac{P_tG_t\sigma}{(4\pi)^2R_t^2R_r^2}$

$$P_r=S_r\cdot\frac{G_r\lambda^2}{4\pi}$$

$$=\frac{P_tG_t\sigma}{(4\pi)^2R_t^2R_r^2}\cdot\frac{G_r\lambda^2}{4\pi}=P_t\cdot\frac{G_tG_r\lambda^2\sigma}{(4\pi)^3R_t^2R_r^2}$$

14.16 $T=\frac{2R}{c}$, 若 $T\le T_r$, 则 $\frac{2R_{max}}{c}\le T_r$, 得 $R_{max}=\frac{cT_r}{2}=\frac{c}{2f_r}$.

14.17 $f_d=\frac{2vf_0}{c}$, $f_0=12GHz$, $v_{min}=1m/s$, $v_{max}=20m/s$.

$$f_{d\,min}=\frac{2v_{min}\cdot f_0}{c}=\frac{2\times1\times12\times10^9}{3\times10^8}=80Hz$$

$$f_{d\,max}=\frac{2v_{max}\cdot f_0}{c}=\frac{2\times20\times12\times10^9}{3\times10^8}=1600Hz$$

the required passband of the Doppler filter is 80~1600Hz.

14.18 $P_t=1kW$, $G=30dB=10^3$, $R=10km$, $\sigma=20m^2$, $\lambda=0.15m$.

$$P_r=P_t\cdot\frac{G^2\lambda^2\sigma}{(4\pi)^3R^4}=2.27\times10^{-11}W=-76dBm$$

$$P_t=10\lg10^3+30=60dBm.$$

$$I=60dBm-(-76dBm)+10dB=146dB.$$

14.19 $P_e=S_t\cdot\frac{\lambda^2G}{4\pi}$, $S_t=\frac{P_e\cdot4\pi}{\lambda^2G}$.

$$P_s=GP_e$$

$$\sigma=\frac{P_s}{S_t}=\frac{GP_e}{P_e\cdot4\pi}\cdot\lambda^2G=\frac{\lambda^2G^2}{4\pi}.$$

14.20 $T_s=(1-\Gamma^2)[\eta_{rad}T_B+(1-\eta_{rad})T_p]$

$$\Delta T_s=(1-\Gamma^2)[\eta_{rad}T_p+(1-\eta_{rad})T_p]-(1-\Gamma^2)[\eta_{rad}T_2+(1-\eta_{rad})T_p]$$

$$=(1-\Gamma^2)\eta_{rad}(T_p-T_2)$$

$$\frac{\Delta T_s}{\Delta T_{true}}=\frac{(1-\Gamma^2)\eta_{rad}(T_p-T_2)}{T_p-T_2}=(1-\Gamma^2)\eta_{rad}$$

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14.21

$R=6400km$, $l=4km$, $\alpha=0.005dB/km$, $T_0=4K$.

$$l=\sqrt{(R+l)^2-R^2}=\sqrt{6404^2-6400^2}=226km$$

$L_{天顶}=0.005\times4=0.02dB\approx1.0046$, $L_{地平线}=0.005\times226=1.13dB\approx1.297$

$$T_e=\frac{T_0}{L}+\frac{L-1}{L}\cdot T_p,\quad T_p=290K.$$

① toward the zenith: $T_e=\frac{T_0}{L}+\frac{L-1}{L}T_0=5.3K$.

② toward the horizon: $T_e=\frac{T_0}{L}+\frac{L-1}{L}T_0=69.6K$.

14.22. (a). $S=\frac{P_tG}{4\pi R^2}$, 则 $R=\sqrt{\frac{P_tG}{4\pi S}}=\sqrt{\frac{5\times1585}{4\pi\times100}}=2.51m$.

(b) $G=32-10=22dB\approx10^{2.2}=158.5$

$$R=\sqrt{\frac{P_tG}{4\pi S}}=\sqrt{\frac{5\times158.5}{4\pi\times100}}=0.8m.$$

(c) $D=\frac{G}{\eta}=2641$, $d=\sqrt{\frac{\lambda^2D}{\pi^2}}$, $\lambda=\frac{c}{f}=0.0107m$.

则 $d=\sqrt{\frac{0.0107\times2641}{\pi^2}}=0.175m$.

$$R_{far}=\frac{2d^2}{\lambda}=\frac{2\times(0.175)^2}{0.0107}=5.7m$$

主瓣和旁瓣的安全距离均小于远场距离，不处于远场区域.

14.23. $S=1300W/m^2$, $\eta_0=377\Omega$.

$$S=\frac{1}{2}|E||H|$$

$$=\frac{1}{2}\cdot|E|^2\cdot\frac{1}{2\eta_0}=\frac{\eta_0|H|^2}{2}$$

$$|E|=\sqrt{2\eta_0S}=\sqrt{2\times377\times1300}=990V/m.$$

$$|H|=\sqrt{\frac{2S}{\eta_0}}=\sqrt{\frac{2\times1300}{377}}=2.6A/m.$$

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